AI-BASED COCONUT DISEASE IDENTIFICATION AND INTELLIGENT ADVISORY SYSTEM

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Project Proposal Report

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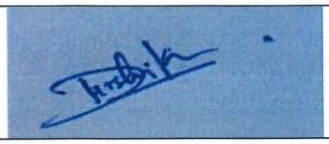
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Coconut cultivation is an important agricultural activity in Sri Lanka, but coconut leaves can be affected by diseases and pest-related conditions that reduce plant health and productivity. Identifying these conditions from images can be challenging for users without immediate access to specialist advice. This project presents an AI-based coconut disease identification and intelligent advisory system combining deep-learning image classification with a Retrieval-Augmented Generation (RAG) chatbot.

The image component uses EfficientNet-B0 to classify uploaded coconut leaf images into four categories: Coconut Caterpillar Infestation (CCI), Gray Leaf Spot, Healthy Leaves, and Leaf Rot. A Mahalanobis-distance-based Out-of-Distribution (OOD) mechanism flags images that may be unfamiliar to the supported classes. The advisory component uses 16 curated agricultural PDF documents. Extracted text is divided into overlapping chunks and embedded using BAAI/bge-large-en-v1.5. The embeddings are stored in Pinecone, where similarity search retrieves five relevant passages for a user query. These passages are supplied as context to GPT-4o-mini through OpenRouter. English and Sinhala interaction and conversation history are also supported.

The system aims to provide a unified interface for initial image-based classification and access to relevant agricultural information. Evaluation considers classification metrics, OOD behavior, retrieval relevance, functional correctness, and usability. It is intended as a decision-support tool, not a replacement for diagnosis or guidance from qualified agricultural officers.

Keywords: Coconut disease identification, EfficientNet-B0, OOD detection, Retrieval-Augmented Generation, Pinecone, agricultural advisory.

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Revise this section to include the correct supervisor names and any domain experts who directly supported the work.

Table of Contents

DECLARATION ..... **Error! Bookmark not defined.**
ABSTRACT 2
ACKNOWLEDGEMENT 3
LIST OF ABBREVIATIONS 6
1. INTRODUCTION 7
1.1 Background and Context 7
1.2 Literature Review 7
1.2.1 Deep Learning for Plant Disease Recognition 7
1.2.2 Deep Learning in Agriculture and EfficientNet 7
1.2.3 Out-of-Distribution Detection 7
1.2.4 Retrieval-Augmented Generation 7
1.2.5 Semantic Embeddings and Vector Search 8
1.2.6 Summary of Related Studies 8
1.3 Research Gap 8
1.4 Research Problem 8
2. OBJECTIVES 9
2.1 Main Objective 9
2.2 Specific Objectives 9
3. METHODOLOGY 9
3.1 Research Design 9
3.2 Dataset and Preparation 9
3.3 Image Classification Model 9
3.4 Out-of-Distribution Detection 10
3.5 Knowledge Base Preparation 10
3.6 Embedding and Retrieval 10
3.7 Response Generation and Language Support 10
3.8 System Integration 10
3.9 Evaluation Plan 10
3.10 Ethics, Privacy, and Security 11
3.11 Limitations 11
4. HIGH-LEVEL SYSTEM ARCHITECTURE 11
4.1 Main Data Flows 12
5. USER REQUIREMENTS 12
5.1 Stakeholders 12

5.2 Functional Requirements 12
5.3 Non-Functional Requirements 13
6. COMMERCIALIZATION PLAN 13
7. BUDGET AND JUSTIFICATION 14
8. WORK BREAKDOWN STRUCTURE 14
10. EXPECTED OUTCOMES 15
11. CONCLUSION 15
12. REFERENCES 15

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
CNN	Convolutional Neural Network
CCI	Coconut Caterpillar Infestation
OOD	Out-of-Distribution
RAG	Retrieval-Augmented Generation
NLP	Natural Language Processing
API	Application Programming Interface
CRI	Coconut Research Institute
SDG	Sustainable Development Goal

1. INTRODUCTION

1.1 Background and Context

Coconut cultivation supports livelihoods, food production, and related industries in Sri Lanka. Coconut palms can be affected by pests and diseases that alter leaf appearance and may reduce plant health if not identified and managed appropriately. Recognizing symptoms early is valuable, although symptoms can vary with condition, severity, lighting, and image quality.

Traditional identification often depends on a farmer's experience or consultation with an agricultural officer. Expert assistance is valuable, but may not always be immediately available. Digital image analysis can provide an initial classification from a photograph, while an advisory interface can help users access relevant management information.

Deep-learning models have been applied to plant disease recognition, but a model trained on a fixed set of categories may still assign an unfamiliar image to one of those categories unless uncertainty is considered. An image label alone also does not explain what a user should do next. Retrieval-Augmented Generation searches a selected knowledge collection and provides relevant passages to a language model as context. This can improve domain relevance but does not guarantee correctness. This project combines a four-class coconut leaf image classifier, an OOD warning mechanism, and a RAG-based advisory chatbot in one web-based system.

1.2 Literature Review

1.2.1 Deep Learning for Plant Disease Recognition

Mohanty, Hughes, and Salathé demonstrated convolutional neural networks for image-based plant disease identification using the PlantVillage dataset [1]. Ferentinos evaluated deep-learning models for plant disease detection and diagnosis across several crops [2]. These studies show the potential of learning visual features from labelled images. However, performance on a research dataset does not guarantee equivalent performance on images captured in different field conditions.

1.2.2 Deep Learning in Agriculture and EfficientNet

Kamilaris and Prenafeta-Boldú reviewed deep-learning applications in agriculture and discussed opportunities and practical challenges [3]. EfficientNet balances network depth, width, and input resolution through compound scaling [4]. EfficientNet-B0 is used here as a transfer-learning backbone adapted to four target classes. Generalization depends on data quality, representative splits, class balance, and careful evaluation.

1.2.3 Out-of-Distribution Detection

A fixed-class classifier generally assumes incoming examples resemble the training distribution. In practical use, people may upload unrelated objects, poor-quality photographs, or leaf conditions outside the supported classes. Lee et al. explored Mahalanobis-distance-based approaches for detecting out-of-distribution samples [5]. This project uses Mahalanobis distance over learned features to flag potentially unfamiliar inputs. It is a warning mechanism and may produce false positives or false negatives.

1.2.4 Retrieval-Augmented Generation

Lewis et al. introduced Retrieval-Augmented Generation, combining a language model with a retrievable knowledge source [6]. A RAG pipeline retrieves relevant passages from a selected document collection and supplies them as context to a language model. This is suitable for a domain-focused advisory tool because

the knowledge collection can be curated and updated separately. The quality of answers still depends on source quality, retrieval relevance, and language-model behavior.

1.2.5 Semantic Embeddings and Vector Search

Sentence embedding models represent text as numerical vectors so semantically related passages can be compared. The BGE embedding family provides models designed for retrieval tasks [7]. Pinecone supports vector storage and similarity search, allowing relevant text chunks to be retrieved for a question before response generation.

1.2.6 Summary of Related Studies

Study	Contribution	Relevance / limitation
Mohanty et al. (2016) [1]	CNN-based plant disease recognition.	Supports image classification; field generalization remains a consideration.
Ferentinos (2018) [2]	Deep-learning models for plant disease diagnosis.	Demonstrates image model potential.
Kamilaris & Prenafeta-Boldú (2018) [3]	Survey of deep learning in agriculture.	Provides context on agricultural AI and deployment.
Tan & Le (2019) [4]	EfficientNet architecture.	Motivates EfficientNet-B0 as the image backbone.
Lee et al. (2018) [5]	OOD detection using feature-distance methods.	Motivates warnings for unfamiliar inputs.
Lewis et al. (2020) [6]	Retrieval-Augmented Generation.	Provides basis for retrieval-grounded advisory.
Xiao et al. (2023) [7]	BGE embedding family and semantic retrieval.	Supports embedding-based knowledge retrieval.

1.3 Research Gap

Image-based plant disease recognition and conversational AI are established research areas, but an agricultural support tool needs to connect visual assessment with accessible domain information. A classifier alone returns a category but may not provide management guidance. A general-purpose chatbot may answer without retrieving from a curated agricultural knowledge base. Fixed-class classifiers may also return a known label for an unfamiliar image unless an explicit OOD mechanism is used.

This project addresses these needs by integrating four-class coconut leaf image classification, Mahalanobis-distance-based OOD warnings, and a RAG chatbot that retrieves passages from curated agricultural documents before generating a response. The system is a decision-support prototype and does not replace expert confirmation.

1.4 Research Problem

Farmers and other users may find it difficult to identify coconut leaf conditions from visual symptoms and obtain relevant management information quickly. Manual identification depends on experience and access to expertise, while an image classifier alone does not necessarily explain next steps. A general language model

may also provide answers insufficiently grounded in domain-specific resources. The research problem is how to integrate coconut leaf image classification, unfamiliar-image warning, and retrieval-based agricultural advisory in one web-based system.

2. OBJECTIVES

2.1 Main Objective

To design, develop, and evaluate an AI-based coconut disease identification and intelligent advisory system that combines four-class leaf image classification, Out-of-Distribution detection, and Retrieval-Augmented Generation to provide an initial visual assessment and relevant agricultural guidance.

2.2 Specific Objectives

1. Prepare and preprocess a labelled image dataset for CCI Caterpillars, Gray Leaf Spot, Healthy Leaves, and Leaf Rot.
2. Develop and fine-tune an EfficientNet-B0 classifier using training and validation data.
3. Evaluate the classifier using accuracy, precision, recall, F1-score, confusion matrix, and loss.
4. Implement Mahalanobis-distance-based OOD detection to flag potentially unfamiliar images.
5. Build a RAG pipeline that extracts text from curated agricultural PDFs, chunks text, creates embeddings, and stores vectors for retrieval.
6. Integrate Pinecone similarity search with GPT-4o-mini through OpenRouter to generate responses using retrieved passages as context.
7. Support English and Sinhala interaction and conversation history as implemented.
8. Evaluate model performance, retrieval relevance, functional correctness, and usability.

3. METHODOLOGY

3.1 Research Design

The project follows a Design Science Research orientation: a technical artefact is designed, implemented, and evaluated to address a practical agricultural information problem. The artefact combines an image-classification service and a retrieval-based advisory service behind a web interface.

3.2 Dataset and Preparation

The image component uses a publicly available coconut disease and pest image dataset. The four target classes are CCI_Caterpillars, Gray Leaf Spot, Healthy_Leaves, and Leaf Rot. Images are resized to the model input dimensions and normalized. Training-time augmentation and class balancing are used to improve robustness and reduce the impact of unequal class counts.

The dataset is split into training, validation, and test subsets using a 70/15/15 split. Related images, near-duplicates, or augmented versions of the same original image should not cross split boundaries because data leakage can produce overly optimistic results. The test set is reserved for final evaluation.

3.3 Image Classification Model

EfficientNet-B0 is used as the feature-extraction backbone. The training strategy has two phases: first, the task-specific classification head is trained while the backbone is frozen; next, a selected portion of the

backbone is unfrozen for fine-tuning. The classification head maps learned features to the four output classes and uses dropout to help reduce overfitting. Validation results guide training decisions.

The model returns a predicted class and class probabilities. Results should be presented as an initial automated assessment rather than a definitive diagnosis. Low-confidence or unfamiliar inputs should be accompanied by a clear caution message.

3.4 Out-of-Distribution Detection

The OOD mechanism uses a 256-dimensional learned feature representation and Mahalanobis distance to estimate how far an input lies from learned class distributions. The service compares the distance with a configured threshold and flags images that may be outside the supported distribution. This approach depends on learned feature statistics and threshold choice; it cannot guarantee that every unfamiliar image will be detected.

3.5 Knowledge Base Preparation

The advisory knowledge base uses 16 curated agricultural PDF documents. Text is extracted and split into chunks of approximately 900 characters with an overlap of approximately 150 characters to preserve context. The current knowledge scope includes coconut caterpillar infestation, coconut leaf rot, and coconut gray leaf spot. Source metadata and scope should be maintained so the collection can be reviewed and updated.

3.6 Embedding and Retrieval

Each chunk is encoded using BAAI/bge-large-en-v1.5, producing 1024-dimensional semantic vectors. The vectors and associated text are stored in Pinecone. When a user submits a question, the system embeds the query and performs cosine-similarity search to retrieve the five most relevant chunks. Retrieval quality depends on the query, chunking, embedding model, and coverage of the indexed documents.

3.7 Response Generation and Language Support

Retrieved passages are supplied to GPT-4o-mini through OpenRouter as context for generating an answer. English and Sinhala interaction support is incorporated through language detection and translation handling. Conversation history supports follow-up questions in saved sessions. RAG reduces reliance on the language model's internal knowledge alone but does not guarantee correctness. If the indexed sources do not support an answer, the system should communicate uncertainty.

3.8 System Integration

The web interface communicates with backend API routes for image prediction and chatbot operations. The prediction service validates and preprocesses the image, invokes the classifier and OOD check, and returns the result. The chatbot service manages sessions, retrieves relevant passages, calls the language model, and returns the response.

3.9 Evaluation Plan

Area	Method	Purpose
Image classification	Accuracy, precision, recall, F1-score, confusion matrix, test loss	Assess overall and per-class performance.
OOD detection	Test supported and unfamiliar examples where available	Review warnings and false positives/negatives.
Retrieval	Manually review passages for	Assess relevance of retrieved

	representative questions	context.
Generated advice	Review relevance, clarity, source alignment, and unsupported claims	Assess usefulness and limitations.
Functional testing	Test upload, prediction, sessions, history, chat, translation, and errors	Verify end-to-end behavior.
Usability	Task-based walkthrough and reviewer feedback	Identify confusing interactions.

3.10 Ethics, Privacy, and Security

Data must be used according to its licence and source conditions. Uploaded images and conversation records should be handled under a clear privacy policy. API credentials should be stored in environment variables and excluded from version control. User-specific sessions should be protected by authentication and authorization. Generated advice should be presented as decision support, particularly for pesticide use or other safety-sensitive recommendations.

3.11 Limitations

- The model only supports the four trained categories and is not a general coconut disease detector.
- Unequal class support may make metrics for underrepresented classes less stable.
- OOD detection can produce false positives and false negatives.
- The chatbot is limited by the coverage and quality of the 16-document knowledge base.
- Generated answers can still contain errors and should not replace expert advice.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The system contains a presentation layer, backend services, an image-analysis pipeline, and a retrieval-based advisory pipeline. These components connect through the web application but perform different tasks.

Module	**Responsibility**
User interface	Upload leaf images, display prediction and warnings, and submit advisory questions.
Prediction API	Validate image files, preprocess inputs, invoke the classifier, and return results.
EfficientNet-B0 classifier	Predict one of the four supported classes.
OOD service	Calculate Mahalanobis distance and flag potentially unfamiliar inputs.
Knowledge ingestion	Extract PDF text, chunk content, create embeddings, and index them in Pinecone.
Retrieval service	Embed a user query and retrieve the five most relevant chunks.

LLM response service	Provide retrieved context to GPT-4o-mini through OpenRouter and return the answer.
Session service	Maintain chat sessions and conversation history.

4.1 Main Data Flows

Flow	Sequence
Image classification	User → Web UI → Prediction API → Preprocessing → EfficientNet-B0 → OOD check → Result and warning → UI
Advisory chatbot	Question → Chat API → Query embedding → Pinecone top-5 search → GPT-4o-mini via OpenRouter → Response → UI
Knowledge ingestion	Curated PDFs → Text extraction → 900-character chunks / 150 overlap → BGE embeddings → Pinecone index

Figure 1: High-level system architecture. Create a visual diagram from these flows for the final submission. Keep the classifier and RAG pipeline distinct, and show retrieval before language-model generation.

5. USER REQUIREMENTS

5.1 Stakeholders

- Farmers and coconut growers who need an initial assessment and access to agricultural guidance.
- Agricultural officers and domain experts who can provide authoritative advice beyond the automated system.
- Researchers and developers who maintain the model, knowledge base, and application.
- Administrators responsible for access control, configuration, and operation.

5.2 Functional Requirements

ID	Requirement
FR-01	The system shall allow users to upload a supported coconut leaf image.
FR-02	The system shall validate image format and file size before prediction.
FR-03	The system shall classify an image into one of the four supported classes.
FR-04	The system shall return the predicted class and probability information.

FR-05	The system shall flag an image when OOD detection indicates it may be unfamiliar.
FR-06	The system shall allow users to ask questions through the advisory chatbot.
FR-07	The chatbot shall retrieve relevant passages before response generation.
FR-08	The system shall support English and Sinhala interaction as implemented.
FR-09	The system shall support chat sessions and conversation history.
FR-10	The system shall show clear errors when requests cannot be completed.

5.3 Non-Functional Requirements

Category	**Requirement**
Usability	The interface should be clear for users with different levels of technical experience.
Performance	Prediction and chatbot requests should complete within a practical time.
Reliability	Invalid uploads, missing configuration, and API failures should be handled gracefully.
Security	Secrets must not be embedded in source code and user data must be protected.
Maintainability	Model configuration, API routes, and ingestion code should be modular.
Scalability	The architecture should support future increases in documents, users, and inference demand.
Transparency	The system should explain that predictions and answers are decision-support outputs.

6. COMMERCIALIZATION PLAN

The intended users include coconut farmers and others seeking an initial understanding of coconut leaf conditions. Agricultural officers and research organizations may also benefit from a consistent first-stage classification tool and access to curated guidance. The system is primarily an academic research and agricultural decision-support prototype.

Future deployment could be explored with agricultural institutions, subject to expert validation, data governance, infrastructure review, and agreement on maintenance responsibilities. Possible delivery models

include institutional deployment, a hosted advisory service, or integration with an agricultural information portal. No institutional partnership or commercial adoption should be claimed unless formally established.

Potential value includes easier access to initial image screening, a unified interface for classification and guidance, and the ability to update advisory content by refreshing the curated documents. Operational costs may include hosting, vector storage, model inference, external language-model API usage, maintenance, and security monitoring. Before real-world use, the system requires field evaluation, expert review, and clear communication of limitations.

7. BUDGET AND JUSTIFICATION

Resource	Estimated cost (LKR)	Justification
Internet and data	4,000	Dataset access, development, and testing.
Cloud hosting	8,000	Prototype hosting and backend availability.
Vector database / API allowance	8,000	Testing retrieval and language-model calls; actual cost depends on usage.
Storage and backups	6,000	Dataset, model, and project backups.
Transport / consultation	6,000	Domain feedback or validation meetings if required.
Documentation / miscellaneous	3,000	Printing, stationery, and incidental expenses.
Total estimated budget	35,000	Indicative estimate; update according to actual expenses and current pricing.

This is an indicative budget, not a statement of actual expenditure. Confirm actual prices and free-tier limits before submission.

8. WORK BREAKDOWN STRUCTURE

Phase	Activities	Deliverables
Initiation	Define problem, scope, users, and success criteria.	Problem statement and scope.
Literature and requirements	Review research and specify requirements.	Literature review and requirements.
Dataset preparation	Inspect labels, split data, preprocess and augment.	Prepared dataset and pipeline.

Model development	Train and fine-tune EfficientNet-B0; evaluate errors.	Model and evaluation report.
OOD implementation	Prepare feature statistics, configure threshold, test warnings.	OOD component and limitations.
Knowledge base	Extract PDF text, chunk content, create embeddings, index vectors.	Searchable knowledge base.
Chatbot integration	Implement retrieval, contextual generation, language support, sessions.	RAG advisory service.
Application integration	Connect frontend, prediction API, chatbot API, and error states.	Integrated web application.
Testing	Run model, functional, retrieval, and usability tests.	Test results and improvement list.
Documentation	Prepare technical documentation, user guidance, and presentation.	Final report and presentation.

10. EXPECTED OUTCOMES

- A web-based prototype that classifies coconut leaf images into four defined categories.
- An OOD warning mechanism to flag potentially unfamiliar images.
- A RAG chatbot that retrieves passages from a curated agricultural knowledge base before generating responses.
- English and Sinhala interaction and saved conversation history, as implemented.
- Evaluation evidence for model performance, retrieval relevance, functional correctness, and limitations.
- Technical documentation of the dataset, model, OOD method, knowledge-base pipeline, and architecture.

11. CONCLUSION

This proposal describes an AI-based coconut disease identification and intelligent advisory system integrating EfficientNet-B0 image classification, Mahalanobis-distance-based OOD warnings, and a RAG chatbot. The image component classifies images into four supported categories, while the advisory component retrieves passages from curated agricultural documents and supplies them to GPT-4o-mini for contextual response generation. Integrating these capabilities may help users move from an initial visual assessment to relevant agricultural information. Practical usefulness depends on evaluation, reliable source documents, transparent communication of limitations, and validation with agricultural experts.

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